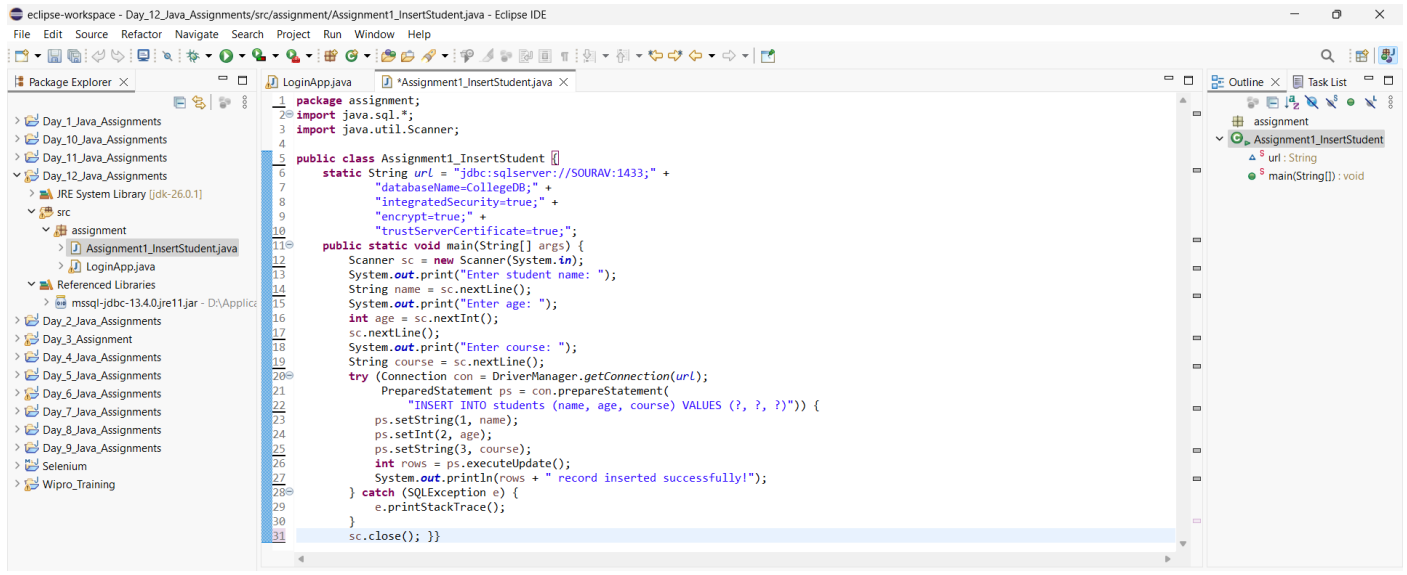


• Create a program to store user input data into database using Scanner and JDBC.



```
1 package assignment;
2 import java.sql.*;
3 import java.util.Scanner;
4
5 public class Assignment1_InsertStudent {
6     static String url = "jdbc:sqlserver://SOURAV:1433;" +
7         "databaseName=CollegeDB;" +
8         "integratedSecurity=true;" +
9         "encrypt=true;" +
10        "trustServerCertificate=true;";
11
12    public static void main(String[] args) {
13        Scanner sc = new Scanner(System.in);
14        System.out.print("Enter student name: ");
15        String name = sc.nextLine();
16        System.out.print("Enter age: ");
17        int age = sc.nextInt();
18        sc.nextLine();
19        System.out.print("Enter course: ");
20        String course = sc.nextLine();
21        try {
22            Connection con = DriverManager.getConnection(url);
23            PreparedStatement ps = con.prepareStatement(
24                "INSERT INTO students (name, age, course) VALUES (?, ?, ?)");
25            ps.setString(1, name);
26            ps.setInt(2, age);
27            ps.setString(3, course);
28            int rows = ps.executeUpdate();
29            System.out.println(rows + " record inserted successfully!");
30        } catch (SQLException e) {
31            e.printStackTrace();
32        }
33        sc.close();
34    }
35 }
```

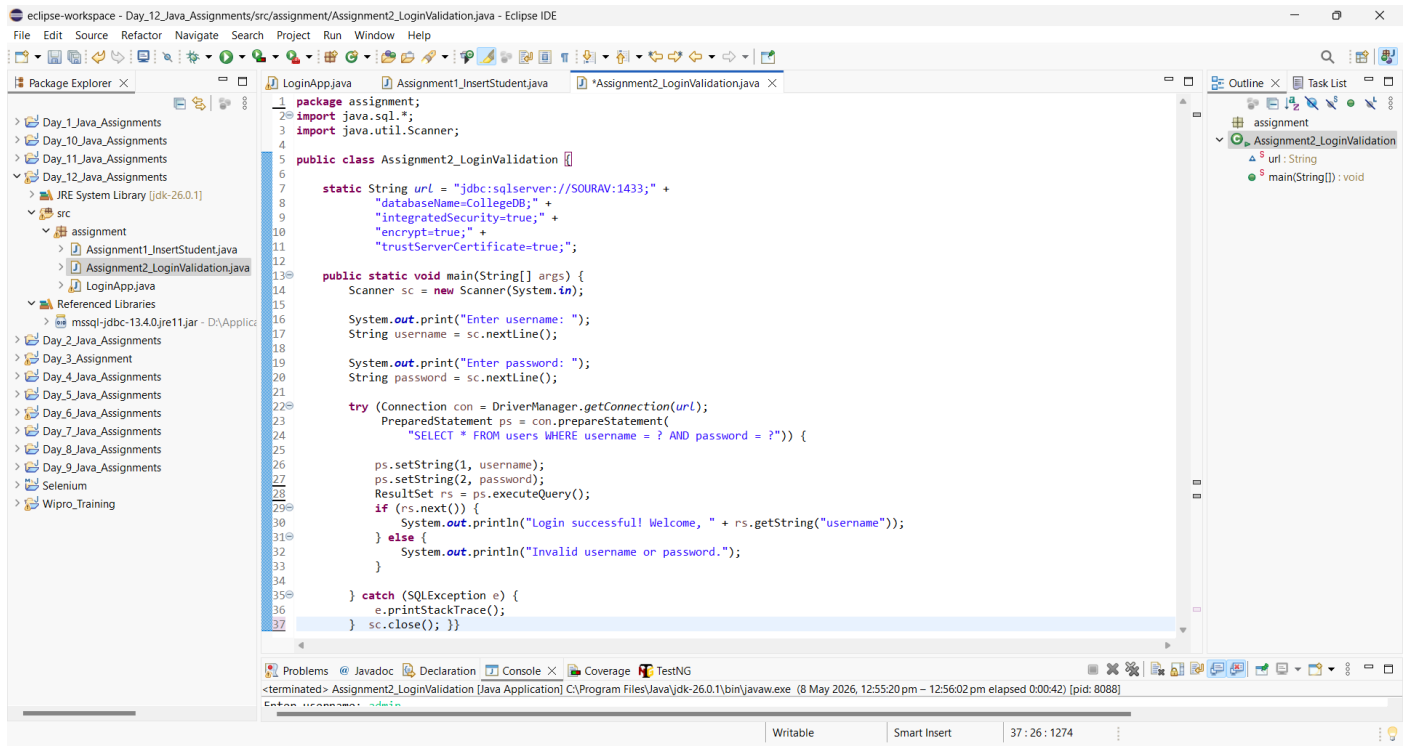
output



```
<terminated> Assignment1_InsertStudent [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (8 May 2026, 12:49:34 pm - 12:51:33 pm elapsed 0:01:59) [pid: 12360]
Enter student name: Sourav Singh
Enter age: 25
Enter course: B.Tech
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by com.microsoft.sqlserver.jdbc.AuthenticationJNI in an unnamed module (file:/D:/Applications/sqljdbc_13.4.0.0_enu/sc
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

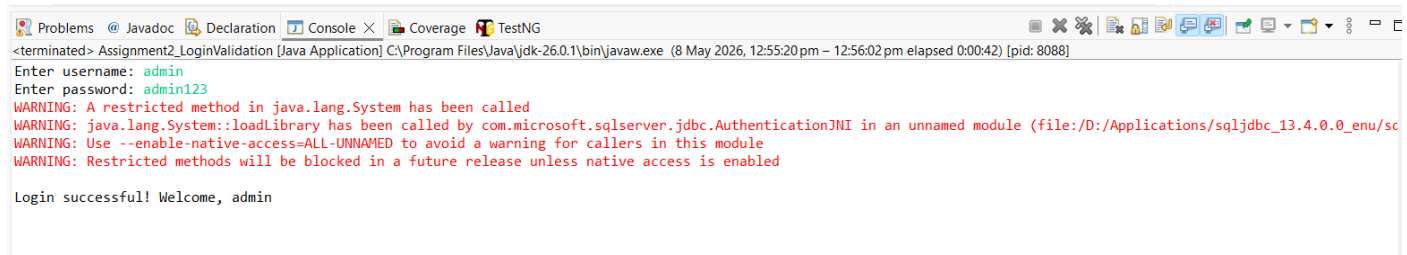
1 record inserted successfully!
```

• Create a program to validate login using username & password from database.



```
1 package assignment;
2 import java.sql.*;
3 import java.util.Scanner;
4
5 public class Assignment2_LoginValidation {
6
7     static String url = "jdbc:sqlserver://SOURAV:1433;" +
8         "databaseName=CollegeDB;" +
9         "integratedSecurity=true;" +
10        "encrypt=true;" +
11        "trustServerCertificate=true;";
12
13    public static void main(String[] args) {
14        Scanner sc = new Scanner(System.in);
15
16        System.out.print("Enter username: ");
17        String username = sc.nextLine();
18
19        System.out.print("Enter password: ");
20        String password = sc.nextLine();
21
22        try (Connection con = DriverManager.getConnection(url);
23             PreparedStatement ps = con.prepareStatement(
24                 "SELECT * FROM users WHERE username = ? AND password = ?")) {
25
26            ps.setString(1, username);
27            ps.setString(2, password);
28            ResultSet rs = ps.executeQuery();
29            if (rs.next()) {
30                System.out.println("Login successful! Welcome, " + rs.getString("username"));
31            } else {
32                System.out.println("Invalid username or password.");
33            }
34        } catch (SQLException e) {
35            e.printStackTrace();
36        } sc.close();
37    }
38 }
```

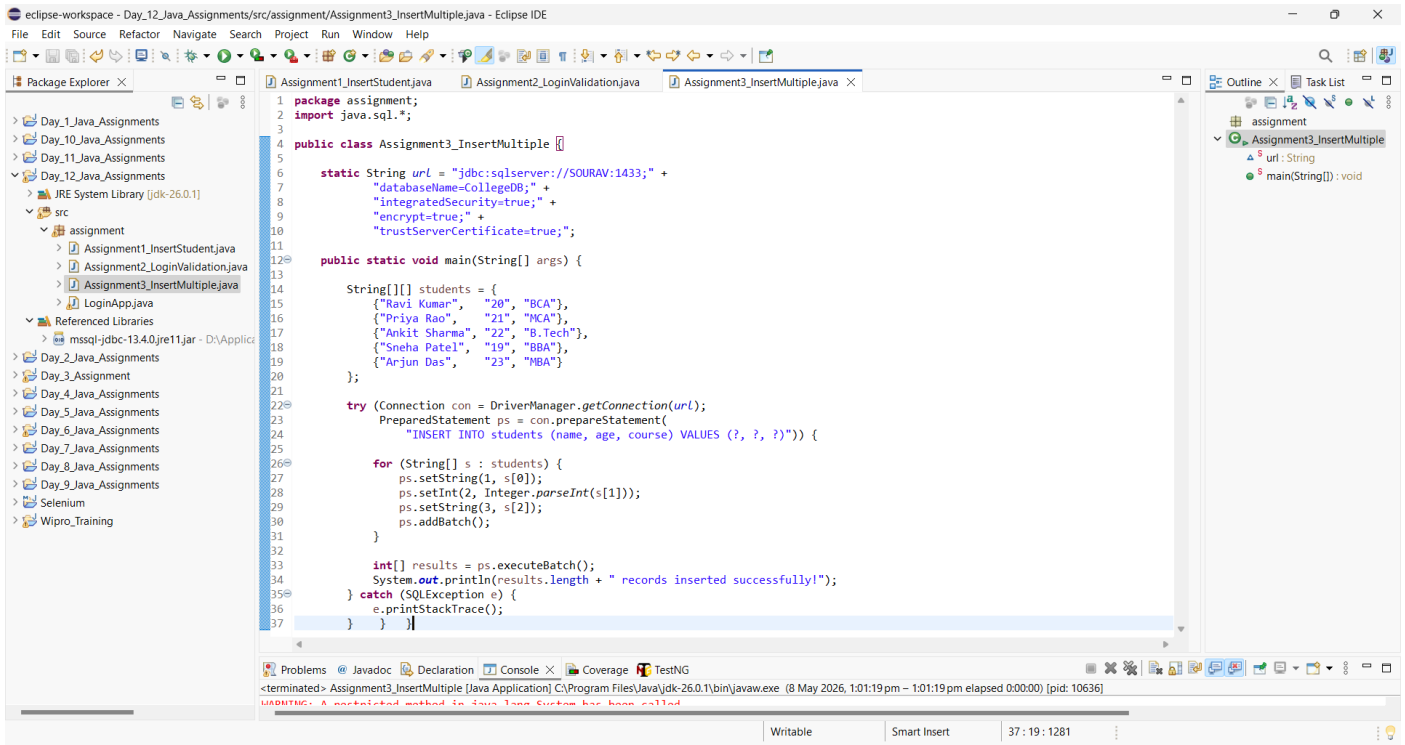
output



```
<terminated> Assignment2_LoginValidation [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (8 May 2026, 12:55:20 pm - 12:56:02 pm elapsed 0:00:42) [pid: 8088]
Enter username: admin
Enter password: admin123
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by com.microsoft.sqlserver.jdbc.AuthenticationJNI in an unnamed module (file:/D:/Applications/sqljdbc_13.4.0.0_enu/sc
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

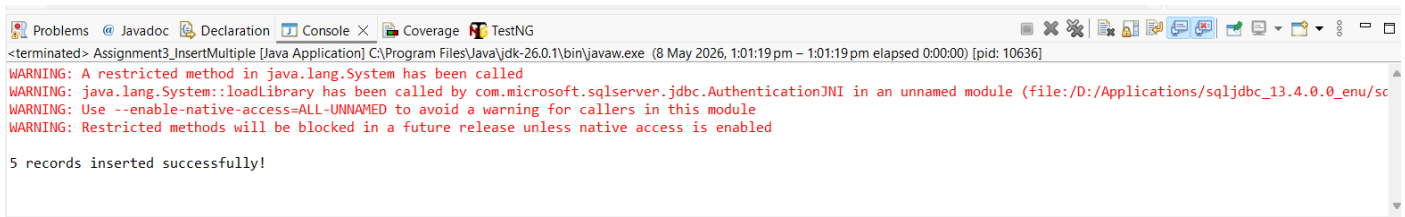
Login successful! Welcome, admin
```

• Write a program to insert multiple records using JDBC.



```
1 package assignment;
2 import java.sql.*;
3
4 public class Assignment3_InsertMultiple {
5
6     static String url = "jdbc:sqlserver://SOURAV:1433;" +
7         "databaseName=CollegeDB;" +
8         "integratedSecurity=true;" +
9         "encrypt=true;" +
10        "trustServerCertificate=true;";
11
12    public static void main(String[] args) {
13
14        String[][] students = {
15            {"Ravi Kumar", "20", "BCA"},
16            {"Priya Rao", "21", "MCA"},
17            {"Ankit Sharma", "22", "B.Tech"},
18            {"Sneha Patel", "19", "BBA"},
19            {"Arjun Das", "23", "MBA"}
20        };
21
22        try (Connection con = DriverManager.getConnection(url);
23             PreparedStatement ps = con.prepareStatement(
24                 "INSERT INTO students (name, age, course) VALUES (?, ?, ?)")) {
25
26            for (String[] s : students) {
27                ps.setString(1, s[0]);
28                ps.setInt(2, Integer.parseInt(s[1]));
29                ps.setString(3, s[2]);
30                ps.addBatch();
31            }
32
33            int[] results = ps.executeBatch();
34            System.out.println(results.length + " records inserted successfully!");
35        } catch (SQLException e) {
36            e.printStackTrace();
37        }
38    }
39 }
```

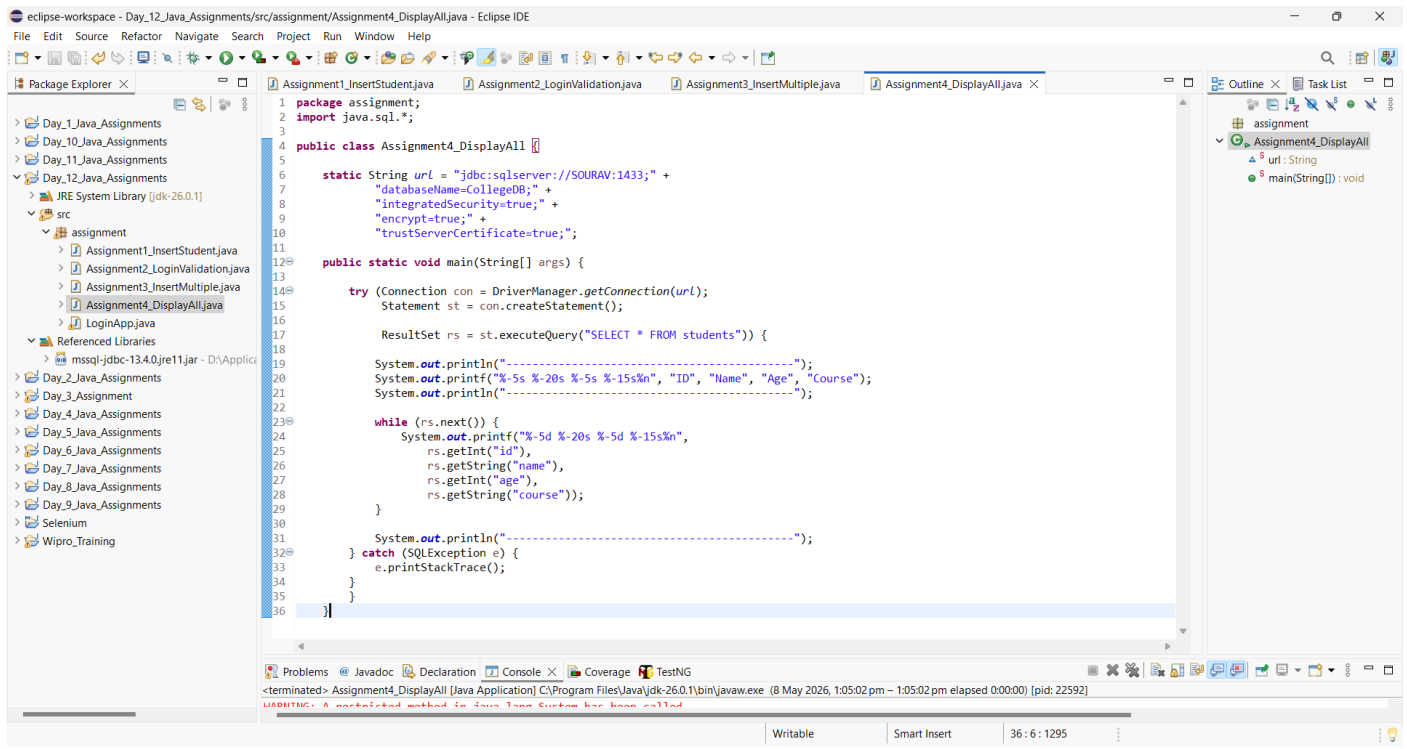
output



```
<terminated> Assignment3_InsertMultiple [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (8 May 2026, 1:01:19 pm - 1:01:19 pm elapsed 0:00:00) [pid: 10636]
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::loadLibrary has been called by com.microsoft.sqlserver.jdbc.AuthenticationJNI in an unnamed module (file:/D:/Applications/sqljdbc_13.4.0_enu/sc
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

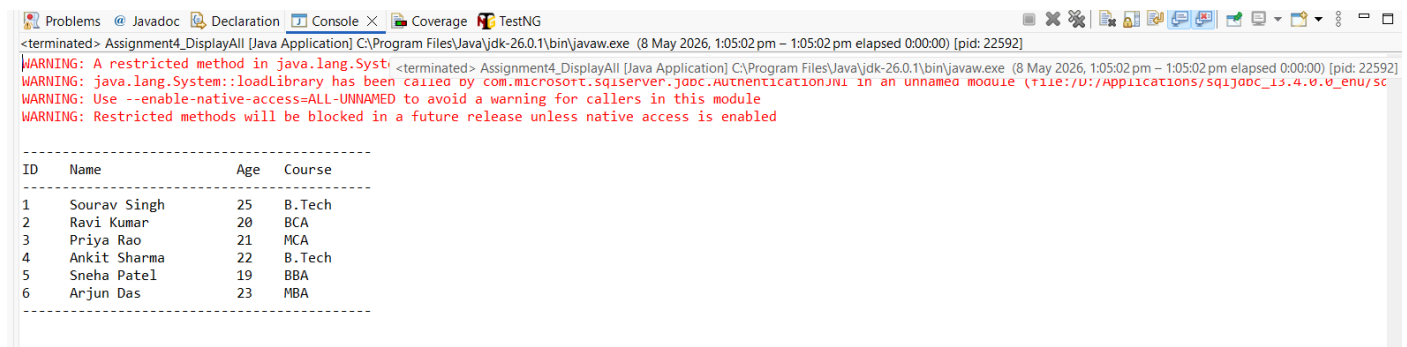
5 records inserted successfully!
```

• Create a program to display all records from students table.



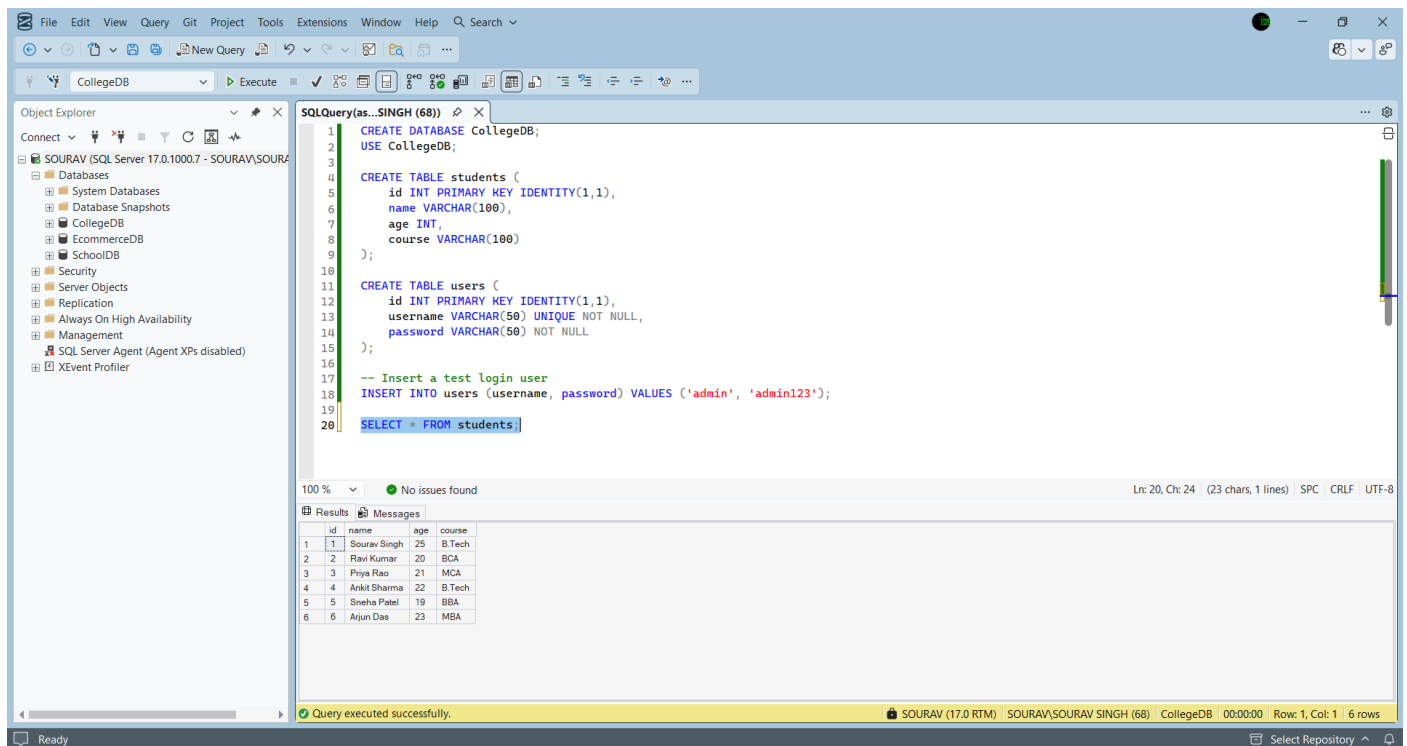
```
1 package assignment;
2 import java.sql.*;
3
4 public class Assignment4_DisplayAll {
5
6     static String url = "jdbc:sqlserver://SOURAV:1433;" +
7         "databaseName=CollegeDB;" +
8         "integratedSecurity=true;" +
9         "encrypt=true;" +
10        "trustServerCertificate=true;";
11
12    public static void main(String[] args) {
13
14        try (Connection con = DriverManager.getConnection(url);
15             Statement st = con.createStatement()) {
16
17            ResultSet rs = st.executeQuery("SELECT * FROM students");
18
19            System.out.println("-----");
20            System.out.printf("%-5s %-20s %-5s %-15s\n", "ID", "Name", "Age", "Course");
21            System.out.println("-----");
22
23            while (rs.next()) {
24                System.out.printf("%-5d %-20s %-5d %-15s\n",
25                                rs.getInt("id"),
26                                rs.getString("name"),
27                                rs.getInt("age"),
28                                rs.getString("course"));
29            }
30
31            System.out.println("-----");
32        } catch (SQLException e) {
33            e.printStackTrace();
34        }
35    }
36}
```

output



```
<terminated> Assignment4_DisplayAll [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (8 May 2026, 1:05:02 pm - 1:05:02 pm elapsed 0:00:00) [pid: 22592]
WARNING: A restricted method in java.lang.System::loadLibrary has been called by com.microsoft.sqlserver.jdbc.AuthenticationJNI in an unnamed module (file://U:/Applications/sqljdbc_15.4.0.0_enu/sc
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

-----
ID      Name                Age  Course
-----
1      Sourav Singh         25   B.Tech
2      Ravi Kumar           20   BCA
3      Priya Rao            21   MCA
4      Ankit Sharma         22   B.Tech
5      Sneha Patel          19   BBA
6      Arjun Das            23   MBA
-----
```



```
1 CREATE DATABASE CollegeDB;
2 USE CollegeDB;
3
4 CREATE TABLE students (
5     id INT PRIMARY KEY IDENTITY(1,1),
6     name VARCHAR(100),
7     age INT,
8     course VARCHAR(100)
9 );
10
11 CREATE TABLE users (
12     id INT PRIMARY KEY IDENTITY(1,1),
13     username VARCHAR(50) UNIQUE NOT NULL,
14     password VARCHAR(50) NOT NULL
15 );
16
17 -- Insert a test login user
18 INSERT INTO users (username, password) VALUES ('admin', 'admin123');
19
20 SELECT * FROM students
```

id	name	age	course
1	Sourav Singh	25	B.Tech
2	Ravi Kumar	20	BCA
3	Priya Rao	21	MCA
4	Ankit Sharma	22	B.Tech
5	Sneha Patel	19	BBA
6	Arjun Das	23	MBA